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COURSE: CHE 312: Transport Phenomena II

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Quiz #1

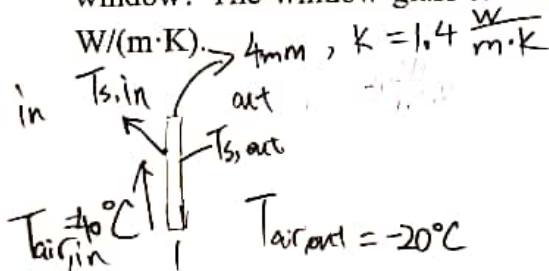
February 5, 2020

Closed Notes and Book

Duration: 20 minutes

06
10

The rear window of an automobile is defogged by passing warm air at 40 °C over its inner surface, and the associate convection coefficient is 30 W/(m²·K). Under conditions for which the outside ambient air temperature is -20 °C and the associated convection coefficient is 65 W/(m²·K), what are the inner and outer surface temperatures of the window? The window glass is 4 mm thick and the thermal conductivity of glass is 1.4 W/(m·K).



$$h_{in} = 30 \frac{W}{m^2 \cdot K}$$

$$h_{out} = 65 \frac{W}{m^2 \cdot K}$$

$$q = \frac{(40 - (-20))k}{\frac{1}{A} \left(\frac{1}{h_{in}} + \frac{L}{k} + \frac{1}{h_{out}} \right)}$$

$$\left[q = \frac{20k}{\frac{1}{A} \left(\frac{1}{30 \frac{W}{m^2 \cdot K}} + \frac{4 \times 10^{-3} m}{1.4 \frac{W}{m \cdot K}} + \frac{1}{65 \frac{W}{m^2 \cdot K}} \right)} \right] \times \frac{1}{A}$$

$$q'' = h_{in} (T_{air,in} - T_{s,in})$$

$$0.0515751 \frac{W}{m^2} = 30 \frac{W}{m^2 \cdot K} (40 - T_{s,in})$$

$$T_{s,in} = 40 - \left(0.0515751 \frac{W}{m^2} \cdot \frac{1}{30 \frac{W}{m^2 \cdot K}} \right)$$

$$= 39.9983^\circ C$$

$$q'' = \frac{q}{A} = \frac{20}{\frac{1}{30} + \frac{4 \times 10^{-3}}{1.4} + \frac{1}{65}}$$

$$= 0.0515751 \frac{W}{m^2}$$

← This changes accordingly

(continue on the back)